INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA

SALTLAKE

(School of University of Engineering and Management, Kolkata)

Degree: BCA, Stream: BCA

Term – II, First Semester Examination, November - 2024

Subject Code: BCACC101

Subject Name: Introduction to C Programming

Full Marks: 30 Duration: 1 Hour

Part - A

Attempt 5 questions

Each question carries 2 marks (2 × 5)

1. Explain function prototype declaration. 2

or

What is recursive function? 2

2. What is the use of strlen()? 2

or

How a content of a string is copied into another string? 2

3. What is the output of the following program:- 2

```
#include <stdio.h>
int main()
{
   int x=5,y=10,z=10;
   x=y==z;
   printf("%d",x);

   return 0;
}
```

or

What is the output of the following program:- 2

```
#include <stdio.h>
int main()
{
   int x=100;
   printf("%d\n",10+ x++);
   printf("%d\n",10+ ++x);
   return 0;
}
```

4. What is a Pointer? 2

or

Describe the statement int *p. 2

5. Define Structure in C. 2

or

Write the syntax of structure. 2

Part - B

Attempt 2 questions

Each question carries 5 marks (5 × 2)

6. Discuss advantages of array over ordinary variable. 5

or

"Array takes contagious memory allocation"---Explain with a proper diagram. 5

7. Differentiate between call by value and call by address with proper example. 5

or

Write a program in C to find the maximum number among three numbers using a pointer. 5

Part - C

Attempt 1 question

Each question carries 10 marks (10 × 1)

8. The library needs a system where the librarian can add, display, search, and delete book records using a simple program. 10
Requirements: Add new books, Display all books, Search for a book by its ID, Delete a book by its ID, Exit the program.
Create a C program to implement the above.

or

A school needs a system to manage student records. The system should allow the administrator to add new students, display all students, search for a student by their ID, and delete a student by their ID. 10

Requirements: Add new students, Display all students, Search for a student by their ID, Delete a student using their ID, Exit the program.
Create a C program to implement the above.

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